
BEVY_ZEROVERSE: AUDITABLE PROCEDURAL INDOOR DATA FOR MULTI-VIEW RECONSTRUCTION

A PREPRINT

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October 1, 2026

ABSTRACT

We describe an open-source synthetic data engine combining primitive-object scenes with furnished procedural interiors, calibrated multi-view cameras, optional human motion, and aligned geometric annotations. Indoor generation samples continuous architectural, furnishing, appearance, lighting and camera parameters under geometric constraints. A continuous camera-baseline control coordinates spacing, shared-surface overlap and path variation. The architectural evaluation audits 512 consecutive rooms and 256 aligned views across 32 rooms. A separately identified camera-baseline reference study covers 2,048 layouts and 512 rendered rooms, with co-visibility measurements at five baseline values. Current capabilities and measured cohorts are distinguished, with exact configurations and reproducible artifacts. Photographic realism, unlimited-process stability and downstream pretraining gains are not established by these diagnostics.

Current native captures · three generated rooms



Figure 1: Current generated architecture: mezzanine stairs, a sunken lounge and a raised platform with an arched portal. Native RGB at $t = 0$; seeds 7, 8 and 6, cameras 3, 3 and 1. Plans, calibrated views, aligned annotations and the unfiltered cohort follow in the main evaluation.

1 Purpose and scope

Sparse-view reconstruction benefits from observations of shared surfaces: cameras that see unrelated objects offer little direct correspondence supervision. LRM-Zero demonstrates the potential of procedurally synthesized object data [1]; MVSpLat uses cross-view matching to estimate geometry [2]. Our contribution is an inspectable data-generation system and its bounded evaluation, rather than a trained reconstruction model or a reconstruction accuracy result.

The engine retains the existing Object, CornellCube, Room, SemanticRoom and Human scene modes, and exposes furnished interiors separately as ProceduralIndoor. The runtime uses Bevy 0.19.1, optional Burn 0.21 inference, and published AnnyBody/ARDY integrations. Office, conference, lounge, classroom, library and related activity priors influence a compositional program; they do not select complete meshes or a finite collection of room templates.

Most architecture, furniture, plants and clutter require neither mesh repositories nor photographic texture collections. AnnyBody human meshes and optional motion/embedding models are explicit pretrained dependencies.

2 Procedural scene program

A scene is generated as $S = G(s, \theta)$, where s seeds independent random streams and θ contains user-controlled densities, activity, quality, camera and motion policies. The serialized manifest records the sampled program, object transforms and seeds, people, photometry and camera paths. A separate generator version and capture-engine identity invalidate incompatible resummptions. Determinism means reproducing the same program and settings with compatible assets and engine versions; bit-identical rasterization across GPU vendors is not promised.

2.1 Architecture, objects and appearance

Room area and aspect, functional zones, aisle widths and activity mixtures are sampled continuously. The exterior footprint supports taper, chamfered corners and a concave cut-in. A shared architectural program controls polygon triangulation, wall apertures, sloping ceilings, arched portals, pillars, floor levels and furnished mezzanines with stairs and guards. Furniture support, camera clearance, fixture anchors and voxel export use the same surfaces. Windows vary aperture dimensions, sill/head heights, bay widths, frames, inset depth and shades across multiple exterior edges. An internal glazed connection retains a furnished neighboring room. This is a bounded architectural grammar: courtyard cut-ins remain on the rear facade, neighboring rooms and mezzanine decks remain rectangular, and arbitrary multi-storey connectivity, structural analysis and building-code compliance are outside scope.

Furniture is assembled from material-specific parts. Tables vary outline, taper, thickness and leg/pedestal construction; chairs vary seat shell, curvature, back proportions, recline, arms, base and headrest. Sectional seating, shelving, potted plants, outlets, switches, ceiling lights, whiteboards, displays and tabletop objects extend the semantic inventory. Placement uses support relationships, clearance checks, zone-dependent furnishing fields and bounded rejection. Object yaw and arrangement are sampled rather than restricted to orthogonal grids. Collision proxies approximate detailed meshes; geometric validity is checked separately from physical plausibility.

Surfaces use 35 role-specific base material slots with procedural albedo, normal and roughness maps, metric UV scales and mip chains. Recipes vary spatial frequency, directional grain, textile structure, mineral veining, relief, wear, color and roughness. Object finish variants preserve their underlying maps while varying appearance; screens, boards and printed pages have independently seeded content. Only referenced finish variants are built during runtime preparation. Glass recipes vary roughness, refractive index, tint, attenuation and thickness. Direct lights vary placement, intensity, color temperature, circuit activity and direction; exposure, daylight and sky radiance are independently recorded. Low-light samples are retained in evaluation.

People retain AnnyBody geometry with continuous phenotype/pose parameters, fitted garments, surface hair and facial details. Static people remain the default. Opt-in ARDY motion combines seeded procedural text sequences with object-aware waypoint planning and scene-time playback. Requesting static scenes does not initialize motion inference. Motion models use the upstream loaders and caches; cached prompt text does not replace the motion seed. Clothing, hair, joint limits and collision handling are approximations, particularly during complex motion.

2.2 Multi-view sampling and training defaults

Camera centers and complete trajectories are constrained to the primary furnished room by default. The sampler varies focal length, position, target, roll and path shape, checks obstacle clearance and prefers visible content. Static capture uses one timestep; optional trajectories support repeated calibrated observations. A reference camera anchors a connected set of views. For each other camera and five trajectory times, we require

$$\min\{o_{0 \rightarrow i}(t), o_{i \rightarrow 0}(t)\} \geq \tau, \quad b_{\min} \leq \|c_i(t) - c_0(t)\| \leq b_{\max},$$

where o estimates first-surface overlap using visibility rays and object/architecture proxies. Overlap constrains reference edges. An additional minimum separation applies to every camera pair at 33 synchronized path samples. Group spread is the horizontal camera-center covariance’s minor/major standard-deviation ratio. Moving pairs must differ in normalized RMS displacement after removing their starting offsets. Full-path swept collision checks remain; sampled spread and overlap are not continuous-time or rendered-pixel guarantees.

The camera-baseline control $b \in [0, 1]$ expands into explicit advanced settings: minimum pair separation $0.04 + 0.46b$ metres, minimum reference distance $0.06 + 2.30b$ metres, maximum reference distance $0.35 + 9.65b$ metres, overlap target $0.65 - 0.55b^{0.6}$, group spread $0.20 + 0.10b$, and independent path variation b . The default is $b = 0.5$. Wide

settings permit distant viewpoints and lower overlap; narrow settings favor close views of shared content. Collision and visibility rejection affect the realized distribution. Arbitrary room, camera-count and constraint combinations need not be feasible.

The default grouped sampler remains biased toward a shared fixation region: peer targets inherit the reference target with only ± 0.35 m horizontal and ± 0.15 m vertical jitter. Their endpoint targets inherit the same offset. Baseline changes positional spacing and path variation but does not broaden this target jitter. Positional diversity and overlap checks therefore do not establish independent gaze or convergence diversity.

The inspector exposes this slider above the advanced controls. Any advanced edit switches to explicit custom values. Changes take effect on regeneration, so dragging a control does not rebuild the room. JSON input accepts `{"baseline": 0.75}` within the `indoor_camera` configuration. Serialization records the expanded values, avoiding hidden preset precedence. Camera travel length and long-route frequency are separate from spacing between cameras. Joint collision and overlap rejection can favor shorter accepted routes at wide spacing; the minimum travel bound remains explicit. The finite search budget reports impossible settings without silently weakening them.

Indoor training configuration	Default and rationale
Capture cameras / timesteps	Four / one; synchronized views of static geometry
Shared-view policy	$b = 0.5$; $\tau \approx 0.287$, pair separation ≥ 0.27 m, reference distance 1.21–5.175 m; explicit <code>multiview:null</code> enables independent views
Group diversity	Horizontal spread ratio ≥ 0.25 for 3+ views; moving-pair normalized RMS difference ≥ 0.075
Room/activity/density	Primary room, Mixed activity, furnishing density 0.65, human density 0.25
Camera path policy	0.03–8 m allowed path length, 0.45 long-path proposal fraction; collision rejection still applies
Human motion	Disabled; static people remain, no ARDY initialization
Native lighting	Auto quality, shadows and approximate baked diffuse GI, 256 rays/probe
Dataset CLI workers / chunks	One process, 16 rooms/chunk; explicit overrides retained
Finite worker lifetime	256 scenes/child for finite indoor jobs with process workers
Default export	RGB and calibration; geometric channels are opt-in

Table 1: Defaults conditional on the procedural-indoor dataset CLI scene. Other scene modes retain their defaults. These are an engineering starting point, not an optimum established by downstream training. Library callers explicitly choose camera and worker counts.

3 Capture and annotation contract

Capture begins only after scene preparation, assets, shader pipelines, lighting and requested human trajectories are ready. A bounded asynchronous readback protocol associates outputs with the requested scene, camera, time and mode; it does not infer readiness from an arbitrary number of frames. Geometric channels share an `RGBA32Float` ground-truth raster attachment. RGB uses Bevy’s PBR pipeline, shadow maps, tone mapping and optional indirect-lighting approximation. `Float32` storage does not imply that all RGB or lighting computations have `float32` intermediate precision: HDR color and irradiance volumes include `float16` stages.

Headless generation supports chunked safetensors or per-sample files, compression, resumable identity checks, metrics and bounded worker replacement. RGB and numeric channels have separate encoding contracts. Native generation performs numeric temporal export and O-Voxel export. The WebGPU viewer supports indoor construction, inspection and optional motion inference, using the same manifest and geometry modules; Portable quality reduces unsupported or costly rendering features. Browser velocity visualization is not a qualification of native dataset flow export. The current Wasm path omits native baked diffuse GI and uses its platform lighting fallback.

4 Current architectural evaluation

4.1 Architectural audit

The current architectural program was audited on 512 consecutive seeds (0–511), with four cameras, furnishing density 0.65 and static human density 0.25. Every room passed layout, support, clearance and camera checks. Oblique walls occurred in 458 rooms, chamfered corners in 307, exterior cut-ins in 221, sloped ceilings in 354, raised floors in 90,

Output	Meaning and qualification
Color	Display RGB; lossless float storage or explicitly chosen codecs. Separate linear-HDR export supports renderer comparisons.
Depth	Colorized, normalized or metric linear depth; axial/ray conventions; calibration records the selection.
Normal / position	View-space normals encoded as $(n + 1)/2$; world positions affine-normalized by the exported AABB, without clamping. First surface, float32 geometric path.
Semantic	Shared class vocabulary, aligned with depth/normal/position, including ceiling light fixtures.
Optical flow	Forward displacement $(\Delta u, \Delta v)$ in pixels, with correspondence-valid and target-visible masks. Terminal time has no forward correspondence.
Motion vectors	The same exported displacement divided by image width/height. Viewer velocity colors are previews, not these numeric annotations.
Co-visibility	Optional same-time, per-pixel membership in other capture cameras. A uint16 mask assigns one bit per camera (up to 16), with a separate source-valid mask and a consistent additive RGB legend; the source camera and editor are excluded.
Calibration / instances	Camera intrinsics, poses and normalized trajectory time, scene manifest, semantic instance oriented boxes, person joints, and primary-room annotation AABB.
O-Voxel	Primary-room architecture/objects only; excludes neighboring/outside geometry. Requires one timestep and disabled human motion; incompatible requests are rejected.

Table 2: Eight currently supported render modes plus structured annotations. The current architectural cohort exports six modes, including production co-visibility. The separately identified reference gallery exports all eight modes, and its population study derives co-visibility independently from depth and calibration. Glass is a first geometric surface: annotations do not follow refracted or reflected light paths. Flow transports surface correspondence, including supported human deformation, rather than estimating photometric image motion.

sunken floors in 103, pillars in 425, arches in 194 and furnished mezzanines in 46. All 512 structural signatures were distinct after quantizing dimensions to 1 cm; this bounded uniqueness does not establish ten-million-sample utility.

The first 32 rooms were rendered without filtering: four cameras at two trajectory endpoints produced 256 views in RGB, depth, normal, semantic, position and co-visibility modes at 640×400 . All views passed annotation alignment. Maximum per-view depth/position p99 discrepancy was $2.87 \mu\text{m}$, reprojection p99 was 0.00264 pixels, and decoded normal-length error was 2.39×10^{-7} . These are annotation consistency checks, not real-world accuracy measurements. The architectural report retains generator-22 identities, complete contact sheets and machine-readable distributions.

Built feature	Rooms	Fraction
Oblique walls	458 / 512	89.5%
Chamfered corners	307 / 512	60.0%
Exterior cut-in	221 / 512	43.2%
Sloping ceiling	354 / 512	69.1%
Raised floor	90 / 512	17.6%
Sunken floor	103 / 512	20.1%
Interior pillars	425 / 512	83.0%
Arched portals	194 / 512	37.9%
Mezzanine	46 / 512	9.0%

Table 3: Architectural features in 512 consecutive rooms. Features co-occur; fractions do not sum to 100%.

4.2 Geometry that is visible and reproducible

Figure 2 pairs native RGB with the same room’s serialized plan and envelope section. A staircase and mezzanine are explicit geometry; the sunken floor has actual risers and support levels. Figure 3 shows a raised platform with an arched portal and a tapered, chamfered outline with an exterior cut-in. These four examples are selected by structural coverage from the first 32 rendered rooms. A feature can be outside a particular camera’s field of view; the project page exposes every camera at both endpoints and the manifest describes the entire envelope. The diagrams are geometric explanations, not additional RGB renders or architectural ground truth from another system.

Footprint areas span $35.05\text{--}262.58 \text{ m}^2$ and occupy 87.3–100% of the axis-aligned bounding rectangle. Ceiling pitch spans $0\text{--}11.32^\circ$, with signed slopes along both horizontal axes. Floor offsets range from -0.506 to $+0.507 \text{ m}$.

Mezzanine decks lie 2.654–3.037 m above the base floor, with 16–18 stair risers and 2–4 solid furniture instances. Figure 4 shows broader shape variation in 12 feature-selected programs from the full audit. These metric constraints are not a building-code certification.

4.3 Aligned images of the current geometry

The main project-page explorer contains all 32 consecutively captured rooms, four cameras and normalized times $t = 0, 1$: 256 camera/time identities and 1,536 displayed channel images. RGB, depth, normals, semantics, positions and co-visibility switch atomically with the corresponding calibration and plan. These are matched six-mode native captures of the current generator; rebuilding the page does not perform a new render. Figure 6 retains pixel alignment and illustrates several different floor levels.

The render configuration is Auto quality, shadows, SSAO and approximate diffuse GI at 1024 primary rays per probe, using native Vulkan on an RTX PRO 6000 Blackwell. RGB display images preserve renderer tone mapping followed by sRGB encoding; no per-image exposure correction, crop or appearance filtering is applied. Depth previews are 8-bit grayscale axial depth divided by 15 m and clamped. Normal previews store $(n + 1)/2$; position previews affine-normalize world coordinates by the annotation AABB and clamp for display. Semantic colors are preserved exactly. Non-RGB PNGs are converted losslessly to WebP, verified by decoded-pixel equality; these 8-bit previews must not be treated as float32 training labels. Original capture checks used the direct float32 ground-truth attachment. Calibration and program downloads retain exact input identities and original PNG hashes.

Every captured view contains at least four semantic classes. All 18 rooms containing people in the camera’s primary functional zone show person pixels in at least one captured view. Across the wider main envelope, 22 rooms contain people; three never show person pixels (seeds 17, 24 and 31). These are class-level observations across both endpoints, not individual-person visibility guarantees. Scene construction through capture/export takes 1.44–4.49 seconds per room (median 2.17), excluding application startup; each room produces eight views in six modes. This is a bounded run, not a sustained-throughput or memory-stability qualification.

4.4 Co-visibility in every architectural room

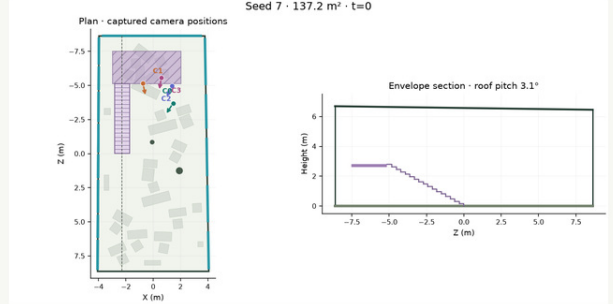
Production co-visibility accompanies RGB and geometric annotations for all 32 rooms, all four cameras and both trajectory endpoints. Each valid source pixel stores an exact 16-bit membership mask of other cameras that see its first surface, with the source bit excluded. One additive RGB legend applies to every view. The unified project-page explorer can show all memberships or highlight an individual peer; the selected peer’s own tile remains the RGB reference.

Across the 256 camera/time views, 83.5% of valid source-pixel observations are visible in at least one other camera. This pixel-weighted statistic uses every room and endpoint; it does not count unique 3D points. Exact downloads contain 16-bit grayscale membership PNGs, separate 8-bit 0/1 validity PNGs, and the matching calibration and legend. The six-mode recapture preserves every serialized room program and the full 512-room parameter distribution of the architectural audit; RGB and annotations are captured together, rather than combining different runs.

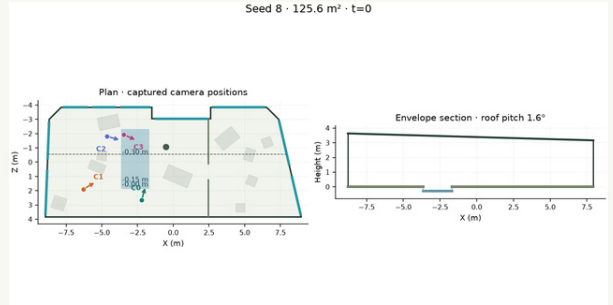
4.5 Population diversity and selection boundaries

Figures 8 and 9 use the same 512-room audit as Table 3. Object counts preserve zero-instance rooms; camera and lighting histograms state their own sample counts. Camera path heatmaps use 33 points per camera. Figure 10 shows camera 0 at $t = 0$ for every rendered room, including dark or sparsely furnished examples. Feature shortcuts do not change any population denominator.

A seed’s bit width does not establish dataset diversity. Continuous parameter ranges have no useful finite cardinality before quantization; topology, placement constraints, correlations and rendering reduce distinguishable outcomes. The 512 unique centimetre-quantized structural signatures exclude seeds, materials and aperture details. They establish bounded uniqueness only. Frozen SigLIP2 embeddings [3] support separate image-space redundancy audits, but no embedding-rank or downstream-learning result is asserted for this architectural cohort.

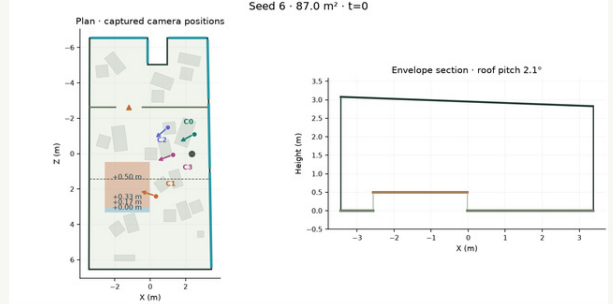
Mezzanine & stairs · seed 7Native RGB · C3 · $t=0$ 

Manifest plan & section · same room

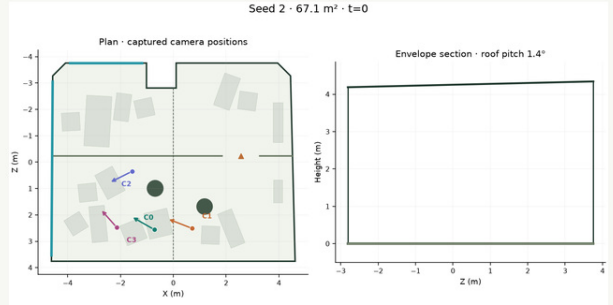
Sunken floor & chamfers · seed 8Native RGB · C3 · $t=0$ 

Manifest plan & section · same room

Figure 2: Current architectural captures with matching plans and sections. Top: seed 7, mezzanine and stair flight. Bottom: seed 8, sunken floor and chamfered, concave envelope. RGB is camera 3 at $t = 0$ in both rooms. Cyan denotes windows; violet marks the deck/stairs; blue marks depressed floors. Gray shapes approximate solid furniture footprints. Camera arrows indicate direction, not visibility frusta; dashed lines locate each section. The envelope sections omit interior furnishings, pillars and partitions.

Raised floor & archway · seed 6Native RGB · C1 · $t=0$ 

Manifest plan & section · same room

Cut-in & tall glazing · seed 2Native RGB · C3 · $t=0$ 

Manifest plan & section · same room

Figure 3: Current architectural captures and their construction programs. Top: seed 6, camera 1, raised platform and curved doorway header. Bottom: seed 2, camera 3, tall glazing in an oblique envelope with chamfers and a rear cut-in. Both at $t = 0$. Orange marks raised floors; triangles mark arched portals in plan. Different features can be visible in different capture cameras. RGB images preserve the complete native view, with no exposure adjustment.

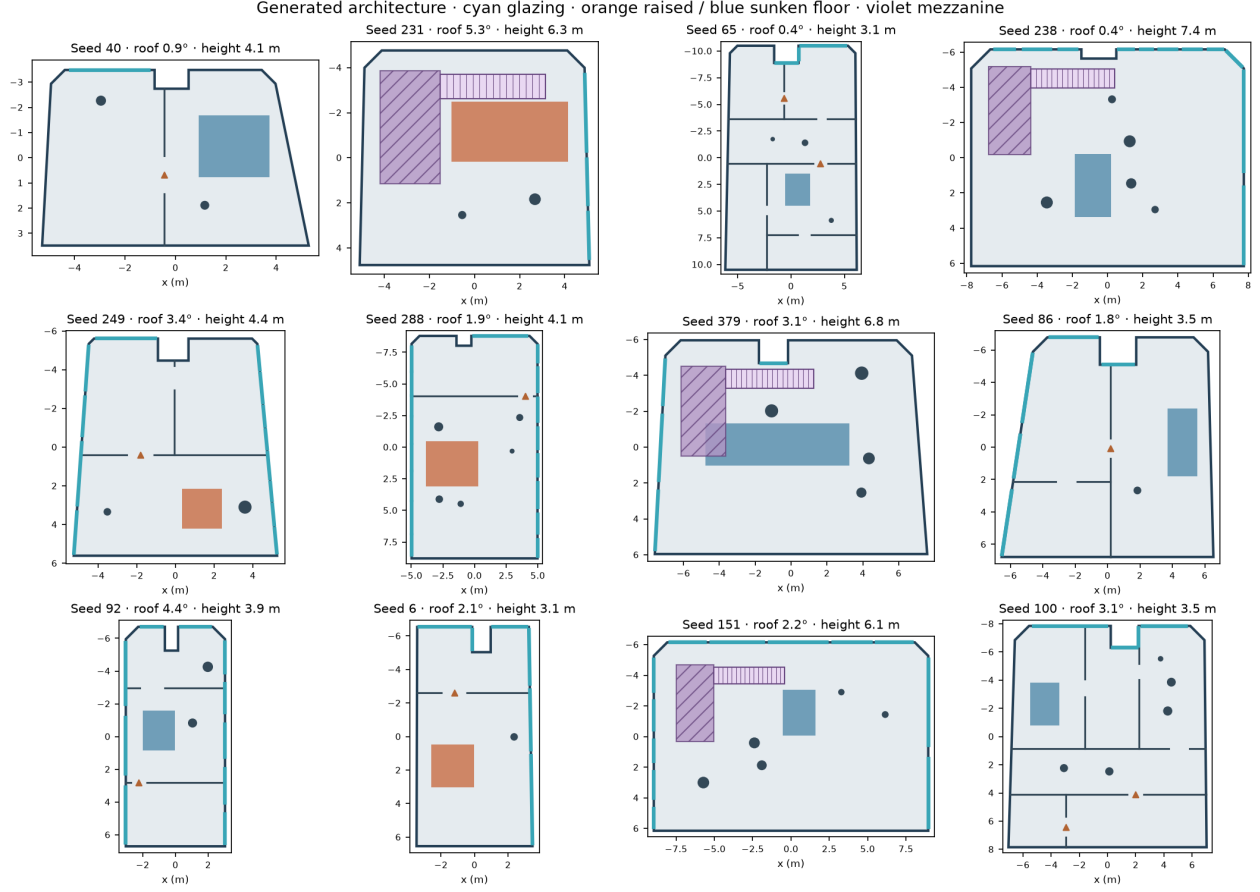


Figure 4: Twelve actual envelope programs selected by architectural feature coverage from 512 consecutive rooms, independently of rendered appearance. Metric coordinates, apertures, floor patches, deck and stair bounds are serialized generation data. Cut-ins and chamfers change the polygon used by construction and clearance checks. Roof pitches and room heights are reported per program. These are diagrams, not twelve additional rendered rooms.

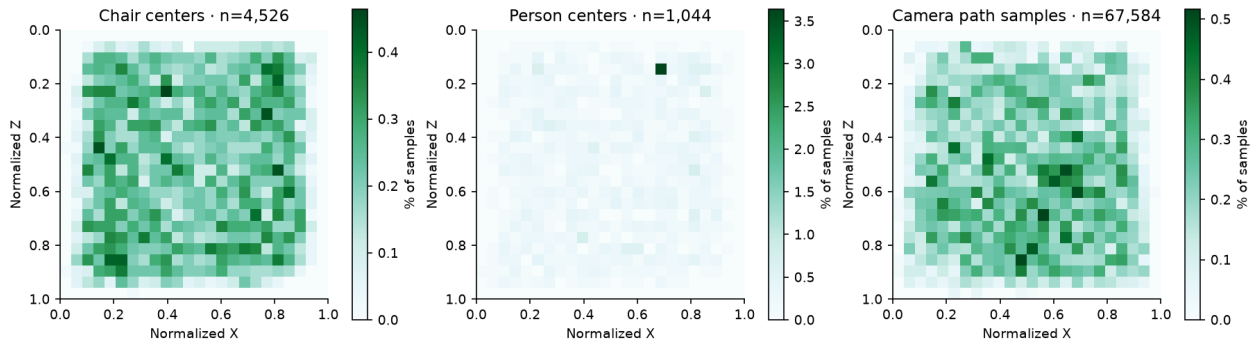


Figure 5: Current population placement in normalized room coordinates: chair and person centers, and 33 path samples per camera. Each panel is normalized independently; +Z points downward. These are locations, not visible pixels or occupied volumes.

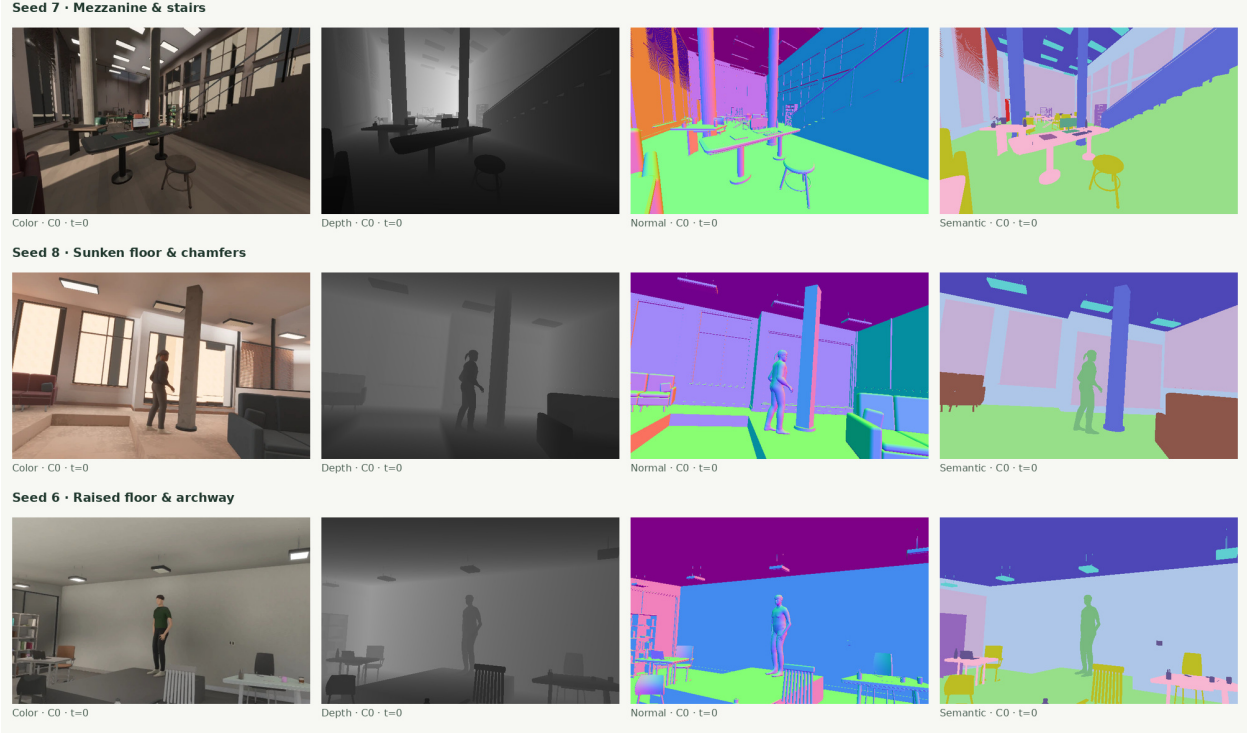


Figure 6: Current captures with matched annotations: RGB, depth, view-space normals and semantics, all from camera 0 at $t = 0$. Rows: seed 7 (mezzanine), seed 8 (sunken floor), seed 6 (raised floor). Full images and the corresponding position previews are accessible in the project-page explorer. Glass is annotation-opaque: RGB visibility through a pane need not match semantic content behind it.

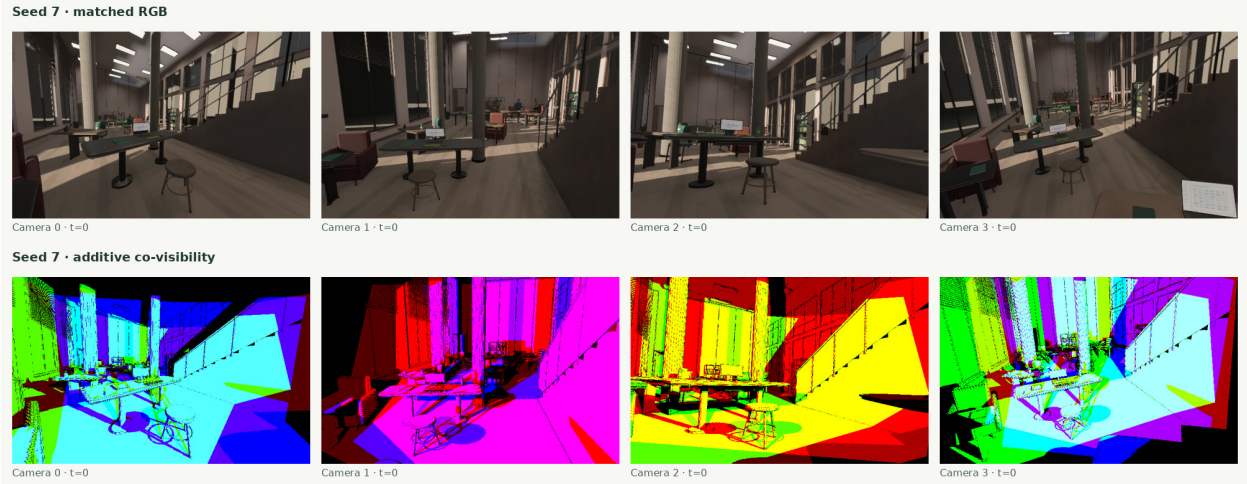


Figure 7: Current architectural scene, seed 7 at $t = 0$: four native RGB views (top) and their production co-visibility annotations (bottom). Columns share the same camera and capture time. All 32 rooms expose this annotation in the same explorer; exact membership and validity downloads preserve the numeric data.

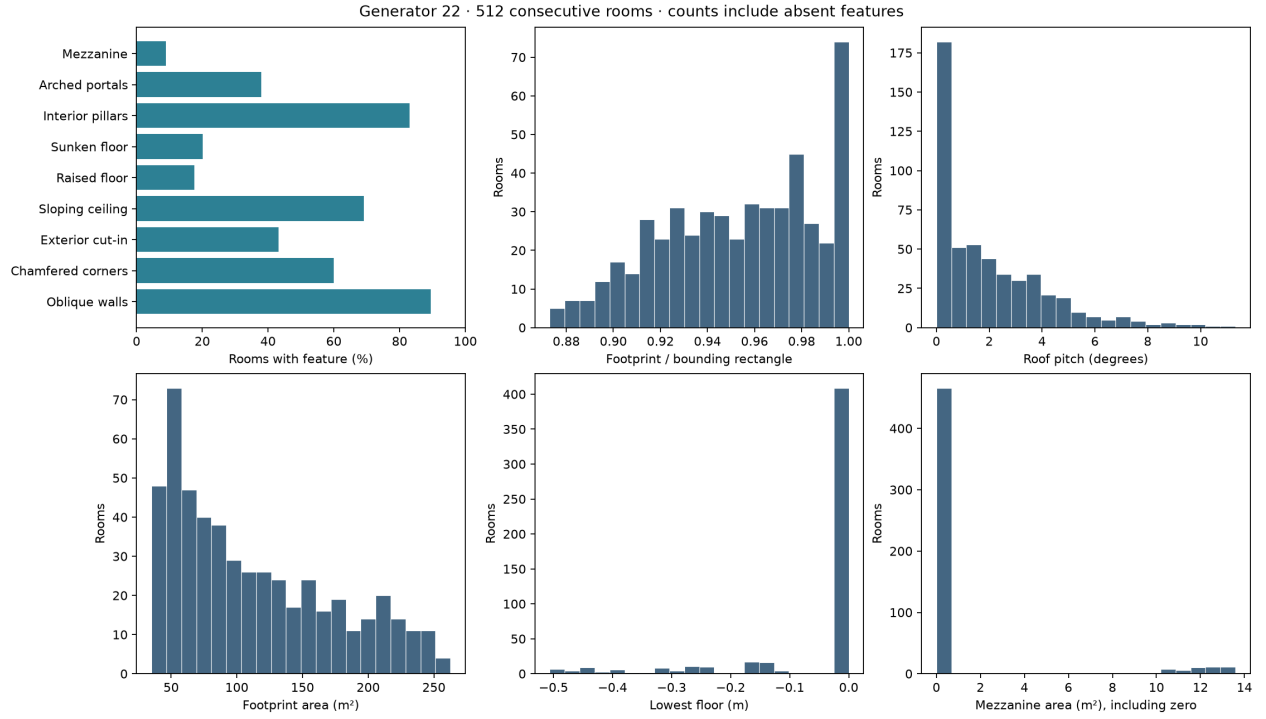


Figure 8: Current architectural distributions across 512 consecutive seeds. Zero-feature rooms are included. Ceiling slope and footprint occupancy measure actual envelope geometry; mezzanine area includes rooms with no deck.

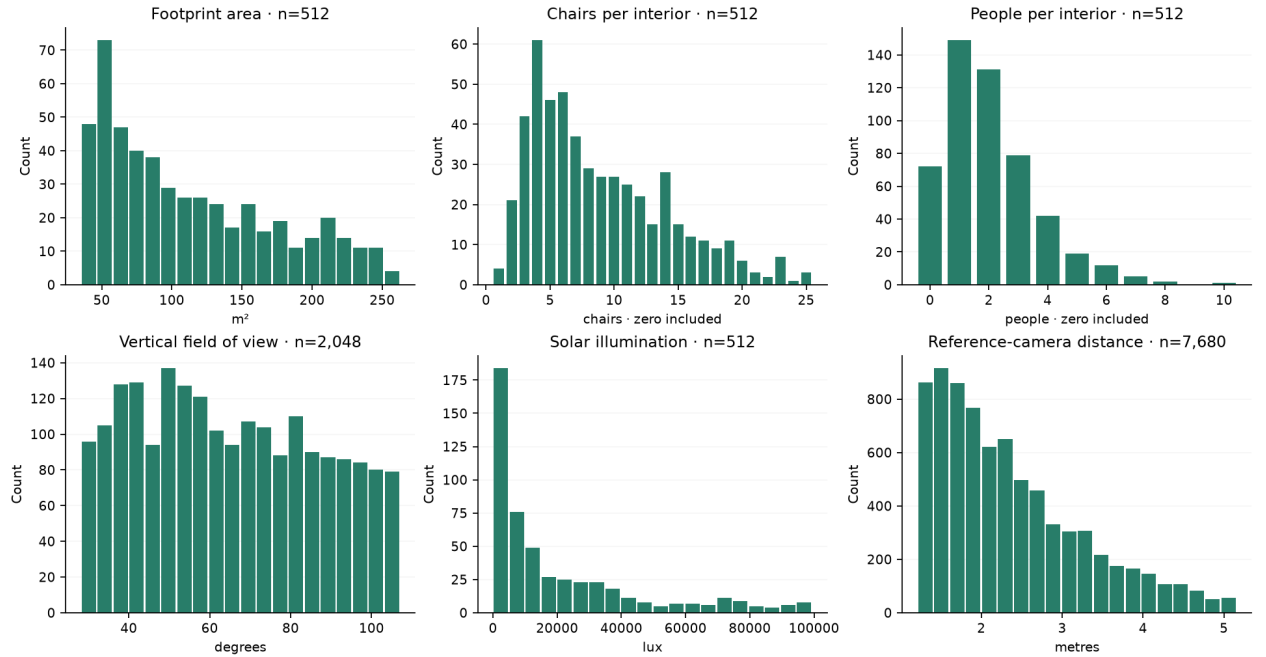


Figure 9: Furnishing, lighting and camera variation in the same architectural population. The footprint measurement uses polygon area, not the bounding rectangle. Every panel states its observation count.



Figure 10: Every consecutively rendered room, seeds 0–31, camera 0 at $t = 0$. No room is removed for appearance. All four cameras, both endpoints and six modes remain available in the current project-page gallery.

5 Limitations and remaining evidence

Procedural correctness, broad parameter variation and efficient capture do not establish photographic realism. Hair is a surface groom, skin lacks a full subsurface model, some furnishings remain sparse, object motion uses approximate collision proxies, and diffuse GI uses probes and simplified transport. Glass and lighting are approximations rather than full path tracing. Physical transport accuracy and visual plausibility require independent path-traced reference studies; the camera-baseline sweep does not measure either.

The current study is a generator audit and a small capture cohort, not a ten-million-scene run. It does not establish long-run resident-memory bounds, balanced coverage of all rare architectural combinations, collision-free arbitrary motion, suitability for every reconstruction loss, or transfer to real photographs. Shared-view defaults deliberately correlate cameras within a room; train/validation splits should therefore be made by room seed, not by image. A downstream assessment should freeze a training budget and real evaluation set, compare matched data quantities and camera policies, and measure reconstruction/calibration quality alongside redundancy. The manifests and metrics make that experiment reproducible without claiming its outcome in advance.

A Recorded camera-baseline reference study

The following baseline, population and gallery results retain their generator-21 capture identities and predate the architectural gallery in the main text. They measure the camera policy in their recorded scenes; they do not remeasure the architectural cohort above. Both studies report absolute measurements with separate denominators. The camera-baseline reference build (generator 21) was audited on 2,048 consecutive layout seeds (24000–26047) at furnishing density 0.65, static human density 0.25, four cameras and default baseline $b = 0.5$. All layouts pass geometry, swept collision and sampled camera constraints. The population contains 508 numeric series, 406 with nonzero observed variance. These are correlated measurements.

The rendered population contains 512 distinct consecutive rooms (24000–24511), four 320×240 cameras and times 0, 0.5, 1. The first 128 rooms are additionally captured at $b = 0, 0.25, 0.75, 1$, yielding 1,024 room/configuration captures and 12,288 views. Every row in the parameter sweep uses the same 128 rooms, with identical geometry, people, materials and lighting, verified by manifest hashes. No failed, dark, or low-overlap sample is replaced. Native Auto lighting and shadows are enabled; baked diffuse GI is disabled for the sweep. The selected gallery uses a separate full-lighting recipe. Captures use the checkout-local wgpu command-cache/upload optimizations; published crates resolve registry wgpu. Timing observations do not qualify unpatched package performance.

Baseline	Distance (m)	Overlap (%)	Shared (%)	All three (%)	Proxy misses
0.00	0.19	72.8	92.1	61.0	177 / 1152
0.25	1.33	55.2	85.2	39.9	96 / 1152
0.50	2.19	46.5	80.7	28.4	52 / 1152
0.75	2.94	38.9	76.9	22.0	22 / 1152
1.00	3.60	32.9	72.8	17.3	11 / 1152

Table 4: Absolute measurements at five settings of the camera-baseline program. Distance is the median room-mean reference distance, measured at 33 times. Overlap averages bidirectional reference pairs at three rendered times. Shared and all-three fractions use valid source-pixel observations. Each row contains 128 rooms, 1,536 views and 1,152 reference pairs; its proxy threshold depends on baseline.

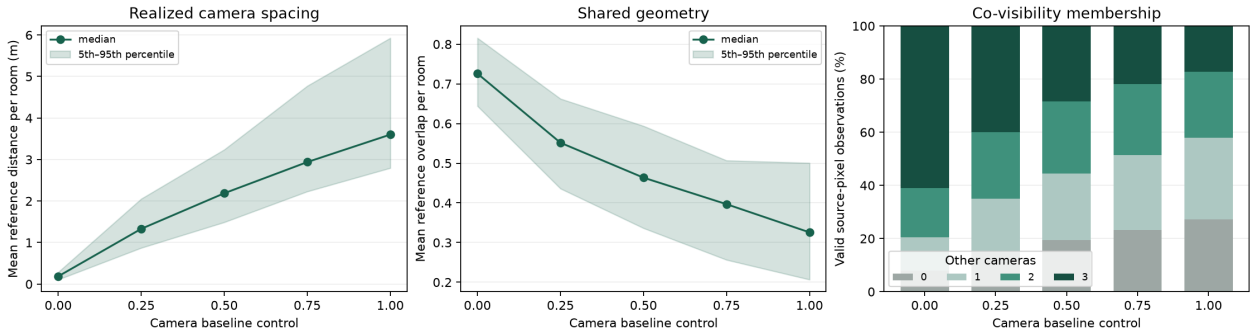


Figure 11: Realized spacing, room-level overlap quantiles and co-visibility cardinality versus the camera-baseline parameter. Pixel observations and views from the same room are correlated.

At default baseline, all 512 rooms contribute 6,144 views and 4,608 reference-pair/time observations. Mean reference overlap is 45.7%, with minimum 16.0%; 210 pairs miss the proxy target. 80.3% of valid source-pixel observations are visible to another camera, and 27.9% to all three others. The denominator is 471,859,200 valid pixel observations, not unique 3D points. Membership excludes the source camera.

The independent diagnostic requires nearest-target-pixel depth agreement within $0.01\text{ m} + 0.002z$. The production GPU annotation uses a separate symmetric tangent-plane predicate; its exact masks are exported with the gallery. Both use the first geometric surface, treating glass as annotation-opaque. Reflections and refractions are not counted as geometric correspondence.

There are 0 rooms with a view containing at most two semantic classes, and 0 of 357 occupied primary rooms have no person pixels in any captured view. Maximum per-view p99 depth/position disagreement is 0.0000038 m. These are annotation and coverage checks, not photographic-realism metrics. Complete distributions, source identities, configurations and commands are in docs/camera_baseline_v21.md. Bounded capture processes do not establish unlimited-process memory stability.

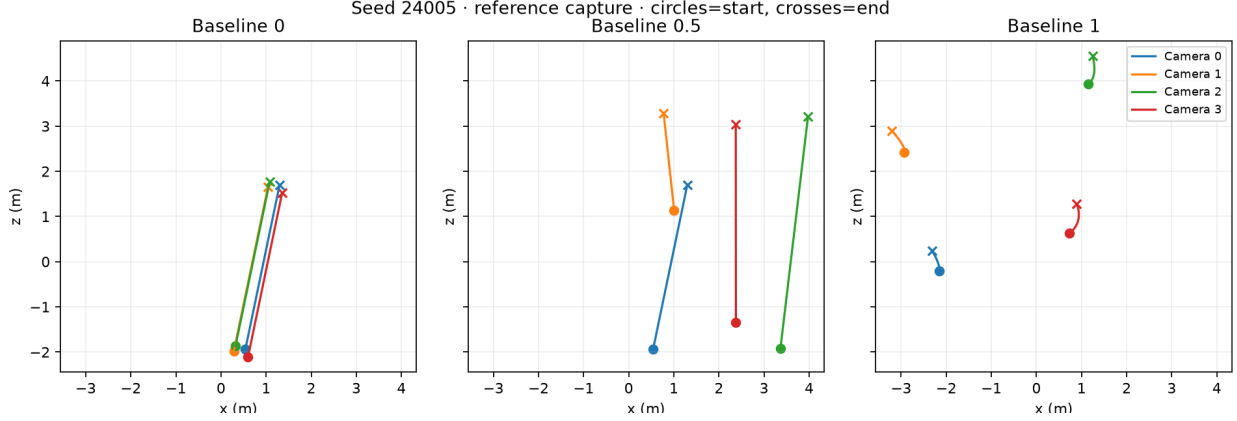


Figure 12: Camera programs for one fixed reference room at three baseline settings, using common metric axes.

A.1 Seeing the baseline control in a fixed room

Figures 13 and 14 show the visual effect of camera-baseline sampling in two fixed scenes. Each column contains all four synchronized RGB views and a top-down camera plan at $t = 0$. Columns change only the camera policy: geometry, furnishings, static people, materials and lighting are identical within a room, verified by hashing all noncamera manifest fields. The project page’s reference gallery provides five captured settings ($b = 0, 0.25, 0.5, 0.75, 1$), an interactive four-view comparison, and a shared-surface overlay backed by exact exported GPU masks. The figures select three of these settings for legibility; they do not select or crop individual camera views.

These selected illustrations use native Auto lighting, 1024 diffuse-GI rays per probe and 768×480 images. They are separate from the consecutive population experiment above. The ten room/configuration captures yield 40 RGB views at one time; the two default groups reuse the matched gallery’s $t = 0$ exports. Image conversion uses a fixed sRGB transfer after renderer tone mapping, with no per-image exposure normalization.

Camera indices identify members of a resampled group, not fixed physical viewpoints across columns. The reference pose and sampled intrinsics may change as rejection finds a feasible group. Hence the examples illustrate the complete baseline policy, not translation of a fixed calibrated rig. Plans retain identical metric axes within each room and show furniture footprints as proxies; short wedges indicate viewing direction, not full frusta or occlusion. Column headers give the mean camera-0 distance at this timestep and the fraction of valid source-pixel observations shared with at least one peer, pooled across all four views. The latter uses the production GPU surface test, distinct from the population depth diagnostic. Pixel observations repeat surfaces; glass remains a first geometric surface.

A.2 Diversity diagnostics

The population export retains zero-preserving object counts, parameter distributions, camera intrinsics and extrinsics, placement heatmaps, correlations and occupancy signatures. These are correlated projections of the procedural program, not independent degrees of freedom. For categorical frequencies p , the entropy-effective count is $\exp(-\sum_j p_j \log p_j)$. The occupancy signature uses a 12-by-12 object-category grid and quantized partition geometry, excluding random seeds and textures. Its HyperLogLog estimate is a bounded geometric redundancy diagnostic, not proof of ten million useful examples.

A seed’s bit width does not establish dataset diversity. Continuous parameter ranges have no useful finite cardinality before quantization; topology, placement constraints, correlations and rendering reduce distinguishable outcomes. Optional frozen SigLIP2 embeddings [3] support image-space redundancy audits, with scene-level aggregation and cross-room nearest neighbors. No embedding-rank or downstream-learning claim is made for this camera-baseline sweep.

B Matched gallery and viewer

The accompanying project page links a separate reference gallery, preserving these earlier captures without presenting them as current architectural examples. Two example rooms (seeds 24005 and 24000) were captured at 768×480 , with four synchronized cameras, three normalized trajectory times and all eight render modes. All mode switches

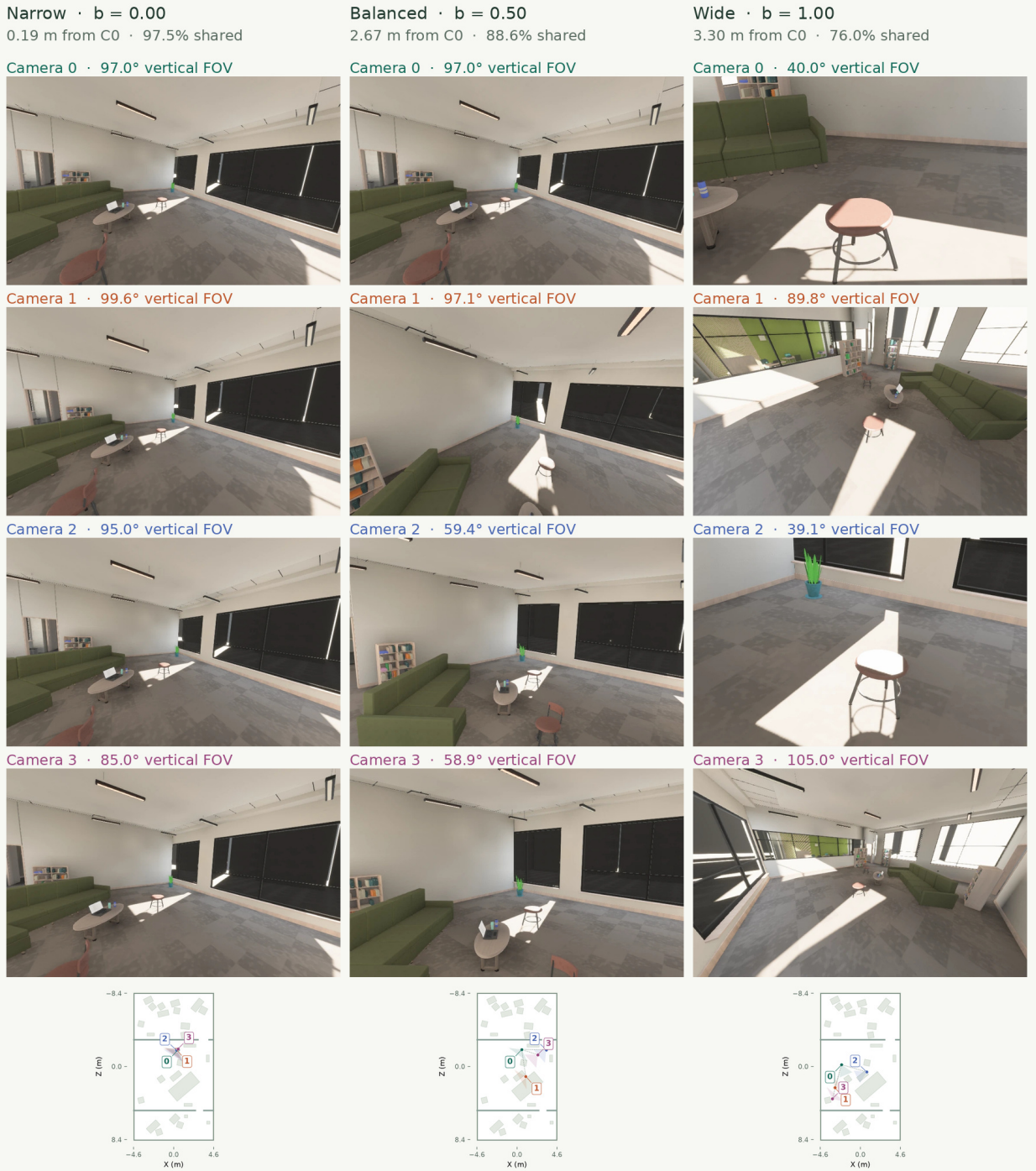


Figure 13: Fixed sunlit seating scene (seed 24005), with narrow, balanced and wide baseline policies from left to right. Rows retain cameras 0–3 in order; the bottom row locates their centers in the same primary room. Close views concentrate on similar content; wider sampling changes perspective and the surfaces jointly observed. Header values describe these particular captures, not population averages.



Figure 14: Fixed workspace scene (seed 24000), using the same column and camera ordering as Figure 13. The room and its lighting are held fixed across baseline values. Feasibility depends on furniture, partitions, camera intrinsics and overlap constraints, so the control is not a requested distance in metres and individual realizations need not vary monotonically.

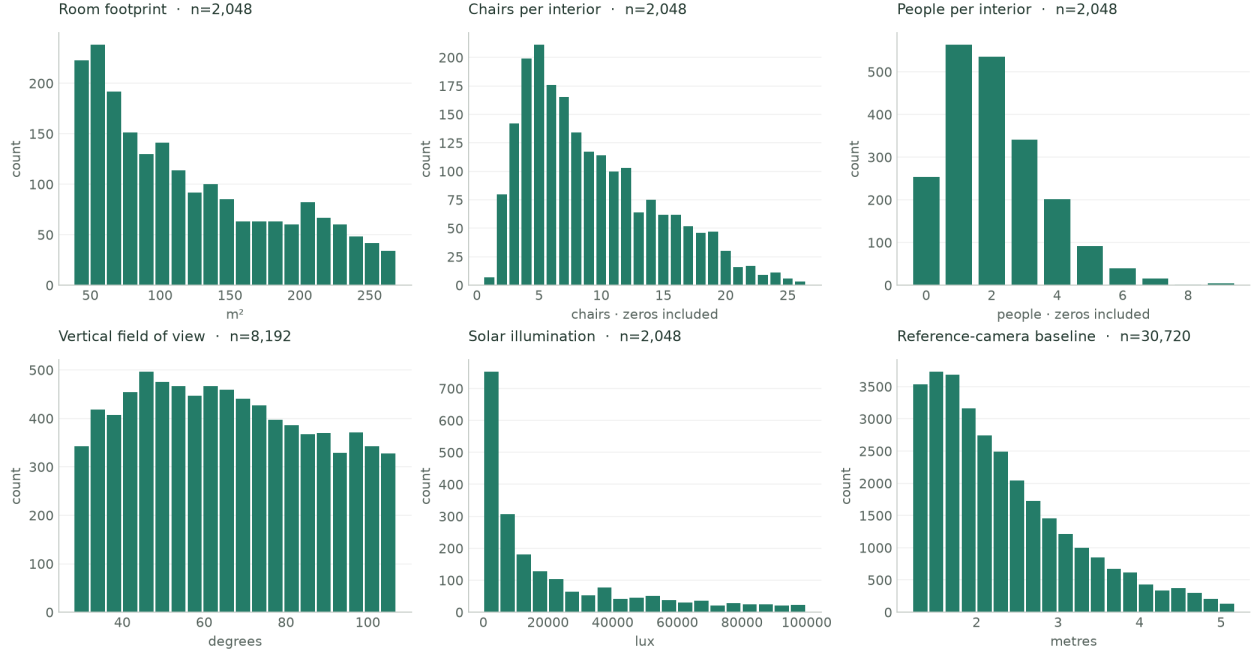


Figure 15: Absolute distributions in the camera-baseline reference population of 2,048 rooms at the default camera baseline. Each panel preserves its denominator and units.

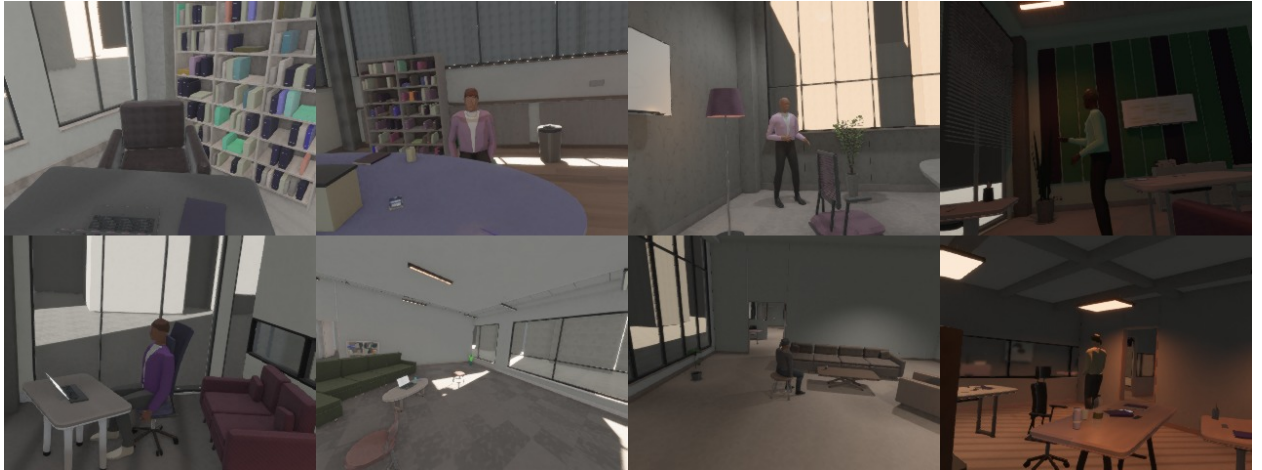


Figure 16: Camera-zero RGB for the first eight consecutive rendered rooms at the default baseline, without aesthetic filtering.

reuse these exact scene/camera/time identities. The native CLI used Auto quality, static human density 0.25 and 1024 indirect-lighting rays per probe. Depth/position reconstruction and camera reprojection checks validate alignment before generating website previews. Display mappings for depth, normals, positions and temporal vectors are declared alongside the visualizations; they do not replace the lossless numeric exports.

B.1 Co-visibility renderings and camera-specific supervision

Figures 18 and 19 show the production GPU annotation for the matched reference gallery. These are actual exported surface-membership masks at the same camera poses, intrinsics and time as the RGB images. They are distinct from the looser depth-based population diagnostic in the baseline table. The figures use the existing selected captures; they add no rooms to the population denominator.

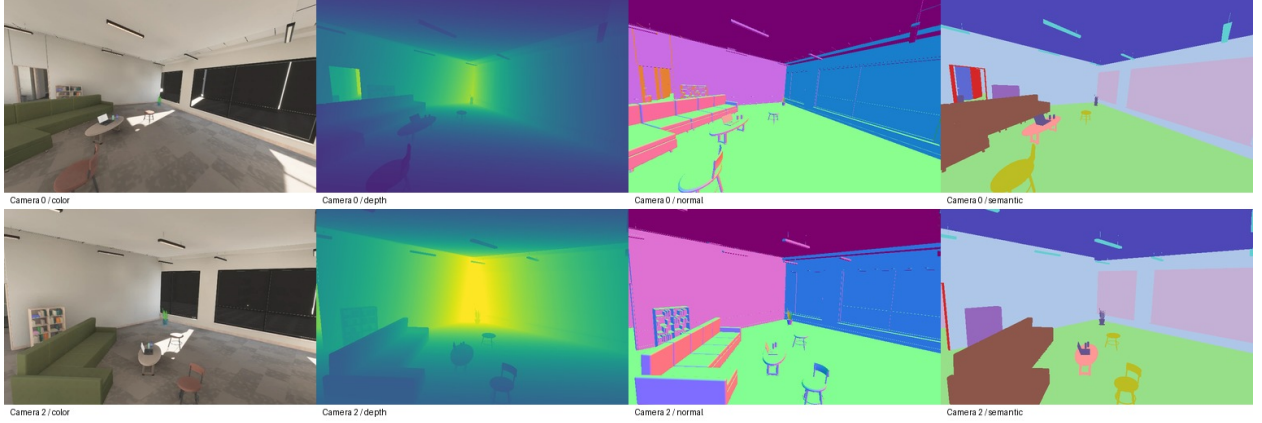


Figure 17: Selected reference room, seed 24005: RGB, metric-depth visualization, view-space normals and semantic labels for cameras 0 and 2 at the same time. Camera selection and pixel-aligned annotation comparisons are interactive on the project page. These illustrations are separate from the consecutive population evaluation.

Co-visibility shares the float32 surface raster used by the other geometric annotations. For source camera s , pixel p and time t , the exported unsigned 16-bit integer is

$$M_s(p, t) = \sum_{\substack{j=0 \\ j \neq s}}^{N-1} 2^j V_{s \rightarrow j}(p, t), \quad 1 \leq N \leq 16,$$

where $V_{s \rightarrow j} \in \{0, 1\}$ records visibility of the source surface in peer camera j . The source camera’s own bit is always zero; the editor has no membership bit. The ordered camera legend remains identical across views and timesteps in a sample. A peer’s binary supervision is recovered by testing $(M_s \& 2^j) \neq 0$; the number of observing peers is $\text{popcount}(M_s)$. Thus the integer preserves camera identities, unlike a shared-pixel count alone.

The GPU pass projects a valid source surface into each peer, checks near/far clipping and image bounds, and reads the nearest target pixel. For source point x , target point y , corresponding world-space unit normals n_s, n_j and target axial depth d , the surface-agreement test is

$$\max(|(y - x) \cdot n_s|, |(y - x) \cdot n_j|) \leq \max(0.001 \text{ m}, 10^{-4} d).$$

This local-plane test avoids comparing depths along different pixel-center rays on sloped surfaces. It remains resolution- and tolerance-dependent: silhouettes, thin geometry, curved surfaces and near-contact surfaces can be ambiguous. Pixel-level membership need not be symmetric after raster reprojection. Glass is the first opaque geometric surface, so reflected, refracted and transmitted RGB content does not imply co-visible geometry.

The additive RGB8 preview sums the contributions of set camera bits without tone mapping. For four cameras, the shared legend is $C_0 = (170, 0, 0)$, $C_1 = (0, 255, 0)$, $C_2 = (0, 0, 255)$ and $C_3 = (85, 0, 0)$. For example, $M = 10 = 0b1010$ means peers 1 and 3, with color $(85, 255, 0)$. These colors encode camera membership, not semantic classes or overlap magnitude. The palette adapts deterministically to camera count and remains uniquely decodable for all supported combinations in the original lossless PNG.

A separate uint8 source-valid mask A_s distinguishes an unshared surface ($A_s = 1, M_s = 0$) from background ($A_s = 0, M_s = 0$); both appear black in the additive preview. Folder exports retain membership and validity as $[H, W, 1]$ arrays in NPZ. Chunked exports use $[B, T, N, H, W, 1]$. Training should consume these numeric arrays and the camera metadata, rather than rasterizing the PDF or decoding a resized or lossy preview. Shared percentages below divide by valid source-pixel observations, including unshared surfaces, not unique 3D points.

All eight $t = 0$ gallery annotations used for this check round-trip exactly between exported uint16 masks and their original additive PNGs. Source validity, self-bit exclusion, shared legends, capture identities and per-peer/count denominators are checked before composing the figures. Original capture hashes and depth/position reprojection are checked separately. The reproducible builder and per-view evidence are `scripts/build_paper_covisibility.py` and `docs/evidence/paper_co_visibility_v21/report.json`.

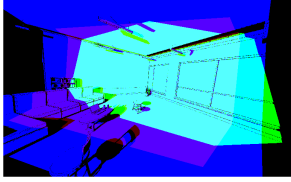
Same room · same timestep · four capture cameras

Camera contribution (RGB8): ■ C0 (170, 0, 0) ■ C1 (0, 255, 0) ■ C2 (0, 0, 255) ■ C3 (85, 0, 0)

Camera 0 · RGB



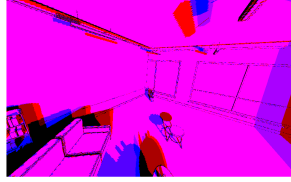
Mask · 80.9% shared



Camera 1 · RGB



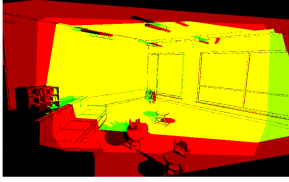
Mask · 95.4% shared



Camera 2 · RGB



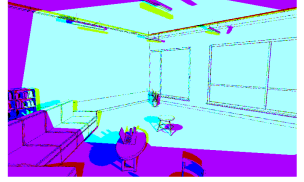
Mask · 79.3% shared



Camera 3 · RGB



Mask · 98.9% shared



Colors add camera membership; they are not semantic classes or a heatmap.

Black = no peer bits. Validity is stored separately. Each source camera excludes its own bit.

Figure 18: Matched RGB (top) and production co-visibility (bottom) for all four cameras in seed 24005 at $t = 0$, from native 768×480 captures. Columns retain the exact viewpoint and full image extent. Camera-bit colors share one legend; each source excludes its own bit. Percentages count valid source pixels visible in at least one peer, for these captures only.

One source pixel, three independent peer-membership bits

Seed 24000 · camera 0 · $t = 0$ · every panel uses the same source-pixel coordinates

Source camera 0 · RGB

Rendered color



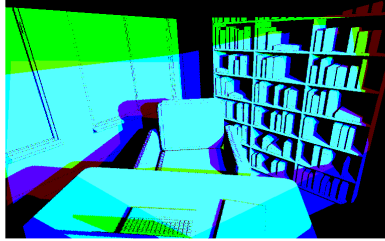
Visible in camera 1

$V1 = (M \& 2) \neq 0$ · 57.8% of valid pixels



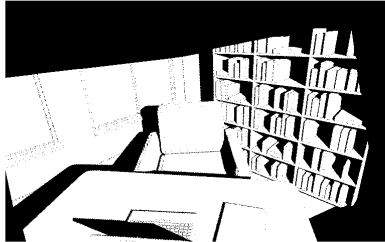
Additive membership color

$M = 2 \cdot V1 + 4 \cdot V2 + 8 \cdot V3$



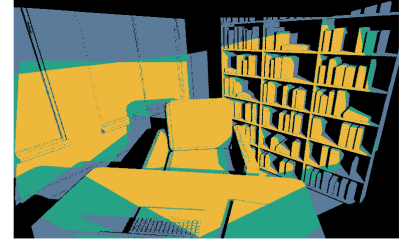
Visible in camera 2

$V2 = (M \& 4) \neq 0$ · 55.3% of valid pixels



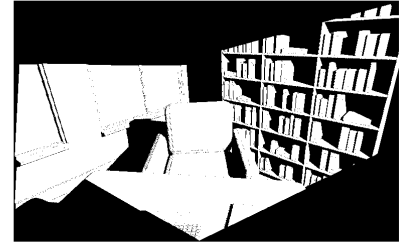
Number of visible peers

$\text{popcount}(M)$, from 0 to 3



Visible in camera 3

$V3 = (M \& 8) \neq 0$ · 44.3% of valid pixels



Peer masks: white = visible in that camera; black = not visible; gray = invalid source.

Peer count: ■ 0 ■ 1 ■ 2 ■ 3 Example: $M = 10 = 0b1010 \rightarrow$ peers C1 and C3

Figure 19: Camera-specific supervision: seed 24000, source camera 0, $t = 0$. Top: RGB, additive code and derived peer count. Bottom: independent peer bit tests, all in source-pixel coordinates. White means visible, black means not visible; invalid source pixels would be gray. The peer masks can overlap: their weighted sum reconstructs membership, and their unweighted sum gives the count. The count palette is separate from the camera legend.

B.2 Human motion illustration

Human clothing uses body-derived cross-sections, localized folds and fitted garment details. Viewer pose joints can show through people while environment depth occludes them; editor projections persist through regeneration. ARDY controls expose diffusion steps, history length, guidance and path conditioning. The reference page’s separately identified motion example contains two accepted trajectories, two cameras and 120 frames at the model’s 20 Hz rate. It is an illustration of the optional motion pipeline, not part of the static camera-baseline experiment or a motion-quality benchmark.

References

- [1] D. Xie et al. LRM-Zero: Training Large Reconstruction Models with Synthesized Data. NeurIPS, 2024. <https://arxiv.org/abs/2406.09371>.
- [2] Y. Chen et al. MVSplat: Efficient 3D Gaussian Splatting from Sparse Multi-View Images. ECCV, 2024. <https://arxiv.org/abs/2403.14627>.
- [3] M. Tschannen et al. SigLIP 2: Multilingual Vision-Language Encoders with Improved Semantic Understanding, Localization, and Dense Features. 2025. <https://arxiv.org/abs/2502.14786>.